A polytope and a triangular hypertope for $\mathrm{Fi}_{22} : 2$

Human Compression project

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Abstract

Explicit permutations of degree 3510 generate the full automorphism group of the Fischer group Fi_{22}. We verify a regular polytope of type $\{6, 13\}$ and, by halving, a regular rank-three hypertope with triangular diagram $(6, 6, 13)$. The finite certificate checks generation, all standard-subgroup intersections and flag transitivity separately. This treats one possible exception in Theorem 5.3 of Piedade–Tranchida; no assertion of priority is made.

1 The explicit group and marking

Let $G =$ `PrimitiveGroup(3510,2)` in GAP's PrimGrp 4.0.1 catalogue. We retain the named identification $G \cong \mathrm{Aut}(\mathrm{Fi}_{22})$ and the classical fact $|\mathrm{Out}(\mathrm{Fi}_{22})| = 2$. The exact installed representation and loader files are hash-bound in `retained_inputs.json`; they remain visible external data, not a new classification proof.

The file `witnesses/Fi22.g` gives three literal degree-3510 permutations $r = (t, a, b)$. This is a complete, deterministic specification: a random seed or an unsuccessful search is not needed for replay. The finite program checks that all three lie in the named group, then independently constructs $P = \langle t, a, b\rangle$ from these permutations and verifies

$$|P| = 129123503308800, \qquad P = G.$$

It computes $H = P'$ and checks that H is simple of order 64561751654400, that $C_P(H) = 1$, that $t \notin H$, and that $a, b \in H$. Thus conjugation on the socle is faithful. The named identification, rather than the numerical order alone, identifies this socle with Fi_{22}.

The three permutations are distinct nonidentity involutions. Their pair-product orders are

$$|ta| = 6, \qquad |tb| = 2, \qquad |ab| = 13. \tag{1}$$

The group equality above is the generation proof used here. For additional validation, the witness file contains words with exact orders 11 and 13. The checker also records all thirteen retained maximal-subgroup orders and confirms that only the socle order is divisible by both primes. With t outer, that classification would give a second generation argument. Completeness of the maximal-subgroup list is not required by the first argument and is not reproved by the additional check.

Put $s = (t, a, a^b)$, where $a^b = b^{-1}ab$. The program checks its pair orders

$$|ta| = 6, \qquad |ta^b| = 6, \qquad |aa^b| = 13. \tag{2}$$

Generation is preserved by the exact identity

$$b = \big(a(a^b)\big)^6 a.$$

Indeed $a(a^b) = (ab)^2$ and $|ab| = 13$, so its sixth power is $(ab)^{12} = (ab)^{-1} = ba$. Direct equality $\langle s \rangle = P$ is checked as well. The identity alone does not establish either incidence condition below.

2 Intersection and flag conditions

For either triple $x = (x_0, x_1, x_2)$ put $G_I = \langle x_i : i \in I \rangle$. Every proper subset of $\{0, 1, 2\}$ is used in the exact test

$$G_I \cap G_J = G_{I \cap J}. \tag{3}$$

There are 49 ordered proper-subset pairs per marking. Each of the other fifteen ordered pairs has a full subset and is an immediate subgroup identity. Thus the full intersection property is established.

Let $P_i = G_{\{0,1,2\} \setminus \{i\}}$. Independently, the checker verifies the three product-set identities

$$(P_j P_k) \cap P_i = (P_j \cap P_i)(P_k \cap P_i), \qquad \{i, j, k\} = \{0, 1, 2\},\ j < k. \tag{4}$$

The reverse inclusion is automatic. For the forward inclusion the program enumerates every $u \in P_j$ and $v \in P_k$, and whenever $uv \in P_i$ tests membership in the right side. No product set is replaced by a generated subgroup. These are 464 pair tests for r and 768 for s.

To identify the geometry, use left cosets with incidence given by nonempty intersection. Normalize a pairwise incident triple so its i-element is P_i. The other cosets have representatives $y, z \in P_i$. Their mutual incidence gives $y^{-1}z \in P_j P_k$. By (4), write $y^{-1}z = uv$ with $u \in P_j \cap P_i$ and $v \in P_k \cap P_i$. The element $yu = zv^{-1}$ belongs to all three cosets. Hence every flag has a common representative and the action is flag-transitive.

Equation (3) identifies all residue stabilizers. Panels have two chambers since each remaining singleton subgroup has order two. Rank-two residues are connected because the corresponding pair generates its dihedral chamber group. Generation of G gives chamber connectedness of the entire geometry. The intersection of all maximal parabolics is trivial, so the chamber action is faithful and regular. A type-preserving automorphism fixing one chamber fixes each uniquely determined adjacent chamber, hence every chamber by connectedness, and therefore every element. Consequently G is the full type-preserving automorphism group.

Theorem 1. *The group* $\mathrm{Fi}_{22} : 2$ *is the automorphism group of a regular polytope of type* $\{6, 13\}$ *and of a regular rank-three hypertope with triangular diagram* $(6, 6, 13)$.

Proof. The explicit finite checks establish the hypotheses of the preceding argument for both markings. For r, the endpoints commute by (1); the standard string-C-group correspondence gives the regular polytope. For s, every pair-product order exceeds two by (2), giving the triangular diagram. Thinness, residual connectedness, regularity and the full type-preserving automorphism assertion have been proved above. □

3 Source and verification scope

Piedade–Tranchida, arXiv:2605.04646v2, Theorem 5.3 on physical page 19, leaves Fi_{22} and Fi'_{24} as possible exceptions to the simultaneous rank-three polytope and nonlinear hypertope assertion. The present result supplies the first case. The second is untouched. This is separate from recovering the source's ten affirmative cases. Neither historical priority nor the absence of other constructions is asserted; a bounded literature audit is separate from the finite mathematical proof.

The complete verifier uses GAP 4.15.1, with PrimGrp 4.0.1 and CTblLib 1.3.11. The literal witness, two full GAP files and complete driver are included. The catalogue representation is retained and hash-bound. Negative controls distinguish a genuine intersection-property marking that fails flag transitivity, wrong generated order, coincident marks, a false halving identity and process errors despite zero exit status. There are no Lean declarations in this packet and no assertion that the exact GAP algorithms have been translated to a proof-assistant kernel. Full external classification-proof costs remain unknown, not zero. No compression factor is claimed.

References

[1] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 28 May 2026.

[2] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[3] R. A. Wilson and collaborators, *Atlas of Finite Group Representations*, `https://brauer.maths.qmul.ac.uk/Atlas/spor/F22/`.

[4] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1; primitive permutation catalogue PrimGrp 4.0.1 and CTblLib 1.3.11.